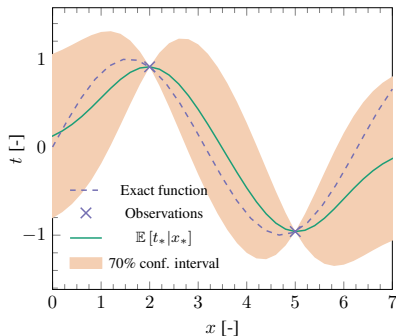


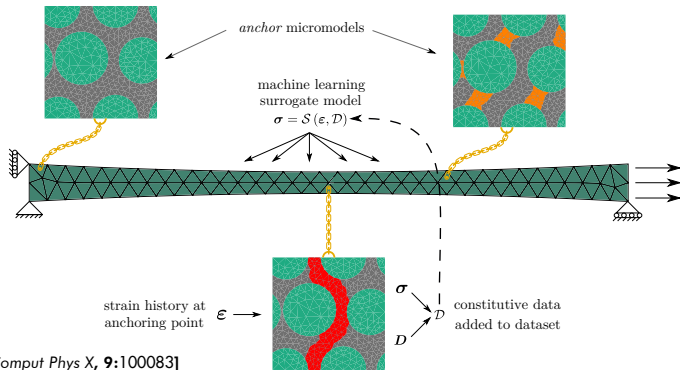
Accelerating FE²: Active learning

Learning constitutive behavior on the fly:

- Macroscopic mesh constrains the region in strain space being approximated
- Eliminates the need for sampling enormous parameter spaces offline



[Rocha et al. (2021), *J Comput Phys X*, 9:100083]



Back to real constitutive models?

Discarding physics has a number of drawbacks:

- Ensuring solution robustness is challenging (negative tangent, spurious localization, etc)
- Non-monotonic strain paths cannot be treated trivially

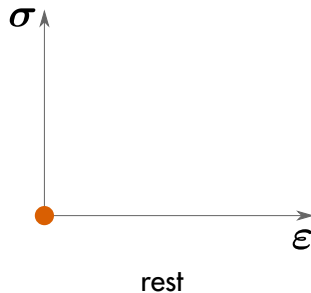
Popular solutions (e.g. RNNs) can be tricky to train:

- Extensive training set needed to guarantee generalization
- Sensitivity to time step size, stopping, etc

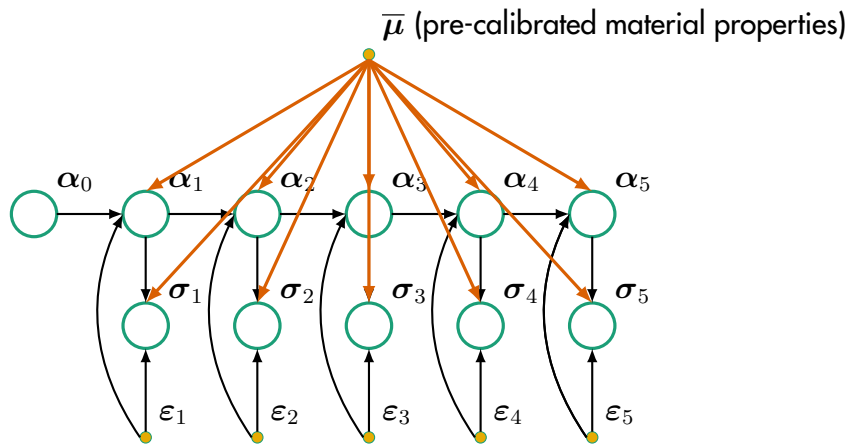
Meanwhile, physics-based models are robust but not expressive enough:

- Constitutive assumptions and fixed pre-calibrated properties

Path dependency, internal variables and material properties



Path dependency, internal variables and material properties



Path dependency, internal variables and material properties

